NYC Taxi Trip Duration Prediction

Exploratory Data Analysis and Model Performance Report

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Project Status: Completed
Model: Ridge Regression

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1. Project Overview

The objective of this project is to predict the duration of taxi trips in New York City using the available trip records.

The project follows a machine learning workflow that includes exploratory data analysis, feature engineering, preprocessing, model experimentation, and final model evaluation.

The prediction target is `trip_duration`, measured in seconds. A `log1p` transformation is applied to the target during training to reduce the influence of large duration values.

The final model uses Ridge Regression with `alpha=1.0`, implemented within a Scikit-learn pipeline.

This report summarizes the main EDA findings, engineered features, model-selection experiments, and final evaluation results.

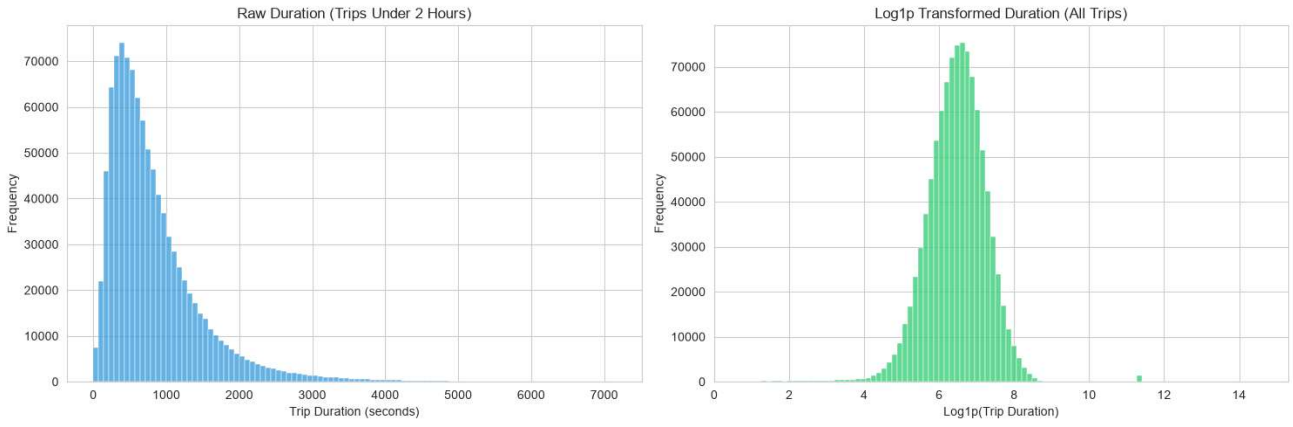
2. Dataset and Exploratory Data Analysis

The training split contains 1,000,000 records and 10 original columns, including pickup timestamps, geographic coordinates, passenger information, vendor identifiers, and trip duration.

The initial data checks reported no missing values or duplicate rows. However, these checks do not guarantee that every observation is valid or geographically plausible.

2.1 Target Distribution

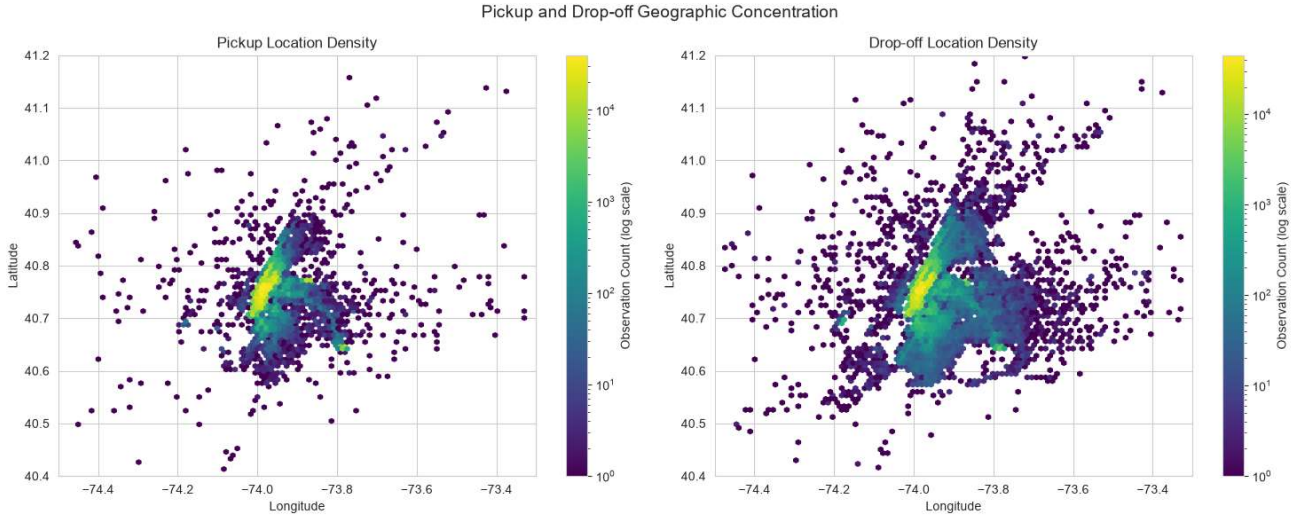
The target distribution was examined on its original scale and after applying `log1p`. The transformation compresses large duration values and provides an alternative target representation for model training.



2.2 Geographic Patterns

Pickup and drop-off coordinates show strong geographic concentration within the selected New York City plotting region.

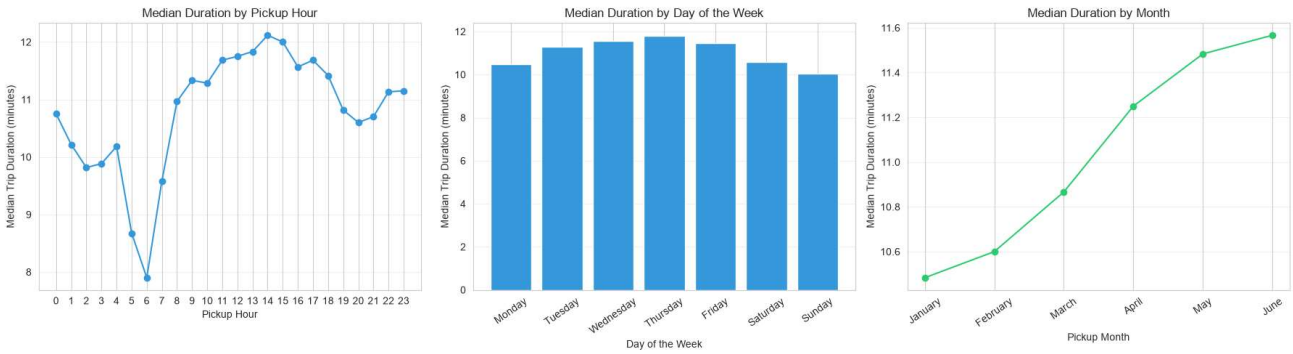
The analysis also identified unusual geographic observations, including trips with identical pickup and drop-off coordinates and extremely large estimated distances.



2.3 Temporal Patterns

Trip-duration distributions vary across pickup hours, days of the week, and the months represented in the training data.

Weekday trips have a median duration of 11.33 minutes, compared with 10.32 minutes for weekend trips. Trips classified as rush-hour trips also have a slightly higher median duration than non-rush-hour trips.



2.4 Distance and Data-Quality Findings

Haversine distance has a median of 2.091 km and a 99th-percentile value of 20.785 km. Manhattan-style distance has a median of 2.686 km and a 99th-percentile value of 28.859 km.

The training data contains 4,085 trips with identical recorded pickup and drop-off coordinates. The maximum recorded Haversine distance is approximately 1,240.909 km, which warrants further investigation in the context of NYC taxi trips.

These observations may indicate unusual records, but the EDA alone does not establish their causes or justify automatically removing them.

3. Feature Engineering

Feature engineering was used to derive additional representations of geographic, temporal, and categorical information from the original trip attributes.

The main engineered feature groups are:

Feature Group	Purpose
Haversine distance	Estimates great-circle distance between trip endpoints.
Manhattan-style distance	Estimates endpoint separation using north–south and east–west components.
Pickup bearing	Represents the initial geographic direction of a trip.
Airport proximity	Measures distances from pickup and drop-off coordinates to JFK, LaGuardia, and Newark reference points.
Temporal features	Extracts pickup hour, weekday, month, and day of the year.
Rush-hour interactions	Combines geographic distances with a rush-hour indicator.
Spatial and day–hour categories	Represents recurring location and time combinations.
Cyclical time features	Uses sine and cosine encodings to represent hourly and weekly periodicity.
Distance relationship features	Calculates the ratio and difference between the two geographic distance estimates.
Log-transformed features	Applies `log1p` to trip duration and selected distance features.

These features provide alternative representations of information already present in the dataset. Their usefulness was assessed through model experiments rather than assumed from EDA observations alone.

4. Model Training and Experiments

Ridge Regression with `alpha=1.0` was used throughout the modeling experiments.

Numerical features were processed using `StandardScaler`, while categorical features were encoded using `OneHotEncoder`. Both transformations were incorporated into a Scikit-learn pipeline.

The initial baseline compared StandardScaler and MinMaxScaler. StandardScaler achieved the higher validation R^2 score and was retained for subsequent experiments.

4.1 Feature Experiment Results

Experiment	Training R^2	Validation R^2
Baseline	0.06436	0.06645
Distance Features	0.35409	0.38955
Distance + Bearing + Airport	0.40152	0.42380
Add Time Interactions	0.40604	0.42549
Add Heavy Hitters	0.68603	0.66113

Experiment	Training R^2	Validation R^2
Add Log Distances	0.72558	0.71050
Add Cyclical Time Features	0.72555	0.71049
Add Distance Relationship Features	0.72825	0.71223

4.2 Spatial Grid Precision

Spatial grid precision was tested to investigate how the level of geographic detail affected generalization.

Grid Precision	Training R^2	Validation R^2
1	0.66529	0.65999
2	0.72825	0.71223
3	0.89245	0.69296
4	0.94799	0.64835

Precision 2 achieved the highest validation R^2 among the tested settings. Higher precision increased training performance but reduced validation performance, indicating weaker generalization.

The final configuration therefore retained the default spatial grid precision of 2 and included the distance relationship features.

The validation split was used to compare model configurations. The test split was reserved for the final evaluation.

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5. Final Model Performance

The final model was trained using the selected feature configuration and saved as `model/ridge_model.pkl`.

Final R^2 Scores

Dataset	R^2 Score
Training	0.72825
Validation	0.71223
Test	0.71629

All reported R^2 scores were calculated against the log-transformed target, `log1p_trip_duration`. They therefore describe performance on the transformed target scale rather than directly measuring prediction errors in seconds.

The final test R^2 of 0.71629 is slightly higher than the validation R^2 of 0.71223. The results indicate broadly similar performance across the validation and test splits.

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6. Conclusion

The project demonstrates that feature engineering substantially improved the Ridge Regression baseline.

Geographic distance features produced a large improvement over the original-feature baseline. Spatial categories and log-transformed distances contributed further improvements, while cyclical time features produced no measurable improvement in the tested configuration. Distance relationship features provided a small additional increase in validation performance.

The selected model achieved a validation R^2 of 0.71223 and a test R^2 of 0.71629 on the log-transformed target.

The EDA also identified unusual geographic observations and zero-distance trips that warrant further investigation. These observations were not automatically classified as invalid solely because they appeared extreme.

Overall, the project combines systematic EDA, engineered geographic and temporal features, controlled validation experiments, and an independent final test evaluation to develop a Ridge Regression pipeline for NYC taxi trip duration prediction.